

Pressure-Sensitive Paint Manufacturing Procedure

This procedure yields ~11mL of Pressure-Sensitive-Paint (PSP).

Materials Required

Safety equipment

- ☐ Fume hood
- ☐ Nitrile gloves
- ☐ Lab coat
- ☐ Safety glasses

Reagents

- ☐ (10 mg) Platinum(II) meso-Tetra(pentafluorophenyl)porphine (PtTFPP)
- ☐ (100 mg) Isobutyl Methacrylate Polymer (polymer)
- ☐ (900 mg) Titanium Dioxide nanoparticles (TiO₂)
- ☐ (10 mL) Acetone

Instruments

- ☐ (1x) Analytical balance
- ☐ (1x) Vial rack
- ☐ (1x) Single vial holder
- ☐ (1x) 20 mL glass vial
- ☐ (1x) 10 mL serological pipette
- ☐ (1x) Pipette bulb
- ☐ (1x) Stainless steel microspatula
- ☐ Weigh paper
- ☐ Labels & lab marker

Procedure

Preparation

1. Gather materials in fume hood.
2. Lay large sheet of weigh paper under work area.
3. Place 20 mL glass vial (mixing vial) into single vial holder.
4. Place PtTFPP vial into appropriate space in vial rack.
5. Place polymer vial into appropriate space in vial rack.
6. Place TiO_2 vial into appropriate space in vial rack.

Add PtTFPP

7. Place mixing vial in holder on balance, uncap, and tare.
8. Uncap PtTFPP stock vial.
9. Transfer 10.0 mg PtTFPP into mixing vial using microspatula.
10. Recap PtTFPP stock vial immediately after transfer.
11. Remove mixing vial in holder from balance.

Add Polymer

12. Cut & place small weigh paper sheet onto balance.
13. Uncap polymer stock vial.
14. Measure 100. mg polymer into weigh paper using microspatula.
15. Recap polymer stock vial immediately after transfer.
16. Add contents of weigh paper into mixing vial.

Dissolve Polymer

17. Uncap acetone stock container.
18. Measure 5.0 mL acetone into 10 mL serological pipette.
19. Add contents of serological pipette into mixing vial.
20. Recap acetone stock container immediately after transfer.
21. Cap mixing vial, mix vigorously until solution appears homogeneous. Uncap mixing vial.

Add TiO_2

22. Cut & place small weigh paper sheet onto balance.
23. Uncap TiO_2 stock vial.
24. Measure 900. mg TiO_2 into weigh paper using microspatula.
25. Recap TiO_2 stock vial immediately after transfer.
26. Add contents of weigh paper into mixing vial.

Final dispersion

27. Uncap acetone stock container.
28. Measure 5.0 mL acetone into 10 mL serological pipette.
29. Add contents of serological pipette into mixing vial.
30. Recap acetone stock container immediately after transfer.
31. Cap mixing vial, mix vigorously until dispersion appears uniform.
32. Label mixing vial "PSP (PtTFPP, Isobutyl Methacrylate Polymer, TiO_2 , Acetone)"

Notes

- If feasible, do not dispose of excess PtTFPP added to the mixing vial. Instead, recalculate other reagent masses as necessary.
- When feasible, keep vials in vial rack to prevent spillage.
- Test batches may be produced by omitting steps under “Add PtTFPP,” for the purposes of experimenting with the manufacturing and application procedures.